

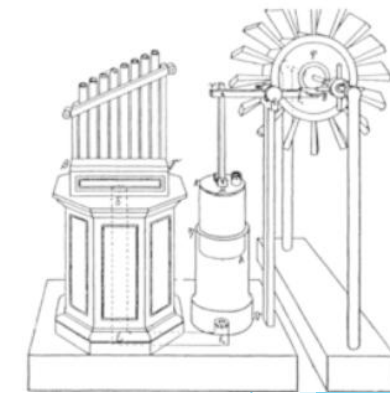
# History of Artificial Intelligence

From Ancient Dreams to Modern Realities

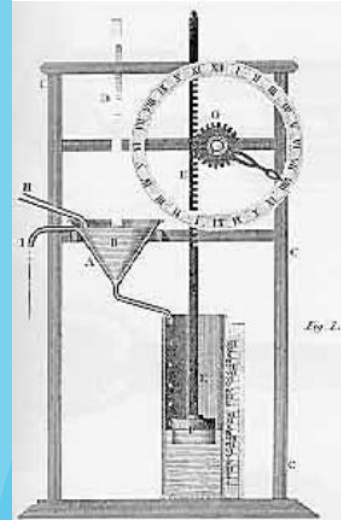
# 1. Early Years (Before 1950s)

## Ancient Foundations

- Early philosophers and mathematicians developed logic and reasoning, laying the foundations for AI.
- Mechanical automata in Greece and later Europe reflected the long-standing human desire to create intelligent machines.



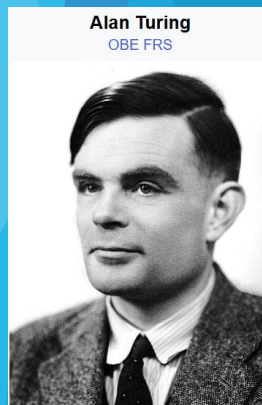
Automata in  
Greek mythology



## Alan Turing's Contributions

- 1936: Proposed the Turing Machine, a model of a general-purpose computing system.
- 1950: Introduced the Turing Test as a way to evaluate whether a machine can display intelligent behaviour.

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- Turing, A.M., 1936. On computable numbers, with an application to the Entscheidungsproblem. Proceedings of the London Mathematical Society, 2(42), pp.230-265.
  - Turing, A.M., 1950. Computing machinery and intelligence. Mind, 59(236), pp.433-460.



Alan Turing  
OBE FRS

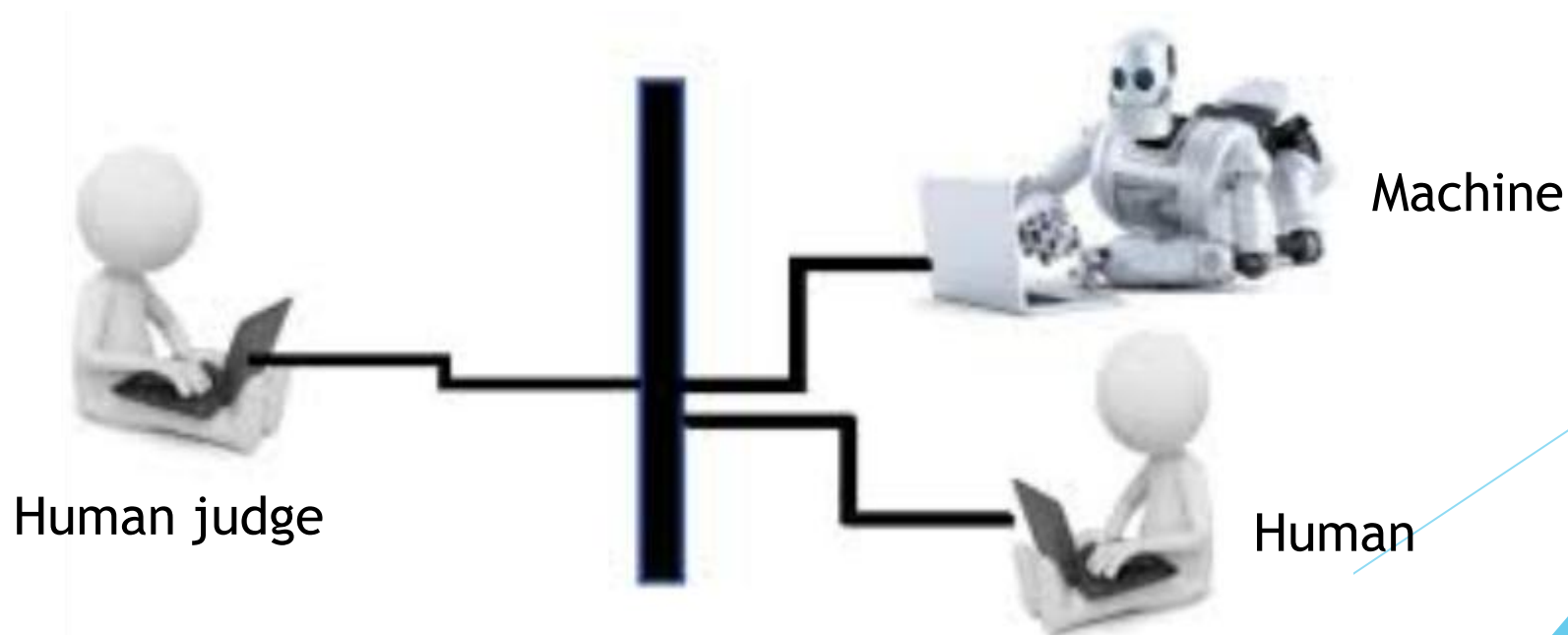
Turing in 1951

# 1. Early Years (Before 1950s)

## Turing Test

A human judge chats with both a human and a machine through text, without knowing which is which.

If the judge cannot reliably tell the machine from the human, the machine is said to pass the Turing Test.



## 2. Golden Years (1950s–1960s)

### The Dartmouth Conference (1956)- Birth of AI

In 1956, scientists from mathematics, computer science, and psychology met at Dartmouth to explore whether human intelligence could be simulated by machines, and formally established artificial intelligence as a research field.

### Early Promising Results

- Logic Theorist (1956) — Often considered the first AI program; it could prove mathematical theorems.
- General Problem Solver (1957-1959) — a research system developed over several years to model human problem-solving using symbolic reasoning.
- ELIZA (1966) — An early natural language program that simulated human conversation.



**Early Pioneers at the  
1956 Dartmouth Conference**

## 3. First AI Winter (1970s)

### Why Progress Stalled

- Expectations were overly optimistic, but technological advances failed to meet them.
- Early systems struggled with real-world complexity
- Computers lacked sufficient computing power at the time
- Government reports criticised the limited practical results
- Funding cuts followed the loss of confidence in AI

## 4. Expert Systems Era (1980s)

After the AI winter, researchers focused on practical applications instead of ambitious human-level intelligence, leading to the rise of expert systems in the 1980s

### Applications

- Medicine — diagnostic systems
- Engineering — design and planning
- Business — decision support

### MYCIN — A Landmark Medical Expert System

- Developed at Stanford University in the early 1970s
- A rule-based expert system for medical diagnosis
- Diagnosed bacterial infections and recommended antibiotics
- Used IF-THEN rules and certainty factors to handle uncertainty

**MYCIN rules may look like:**

```
(rule R1
  (if
    (and
      (infection primary-bacteremia)
      (culture-site sterile-site)
      (portal-of-entry
        gastrointestinal-tract))
    )
  (then
    (organism bacteroides)
    (certainty 0.7)
  )
)
```

## 5. Second AI Winter (Late 1980s–Early 1990s)

### Why Expert Systems Lost Popularity

- Expensive to build, maintain, and update
- Could not handle situations outside their narrow rule sets
- Knowledge acquisition from humans was slow and costly
- Knowledge could only be updated manually; systems could not learn from data

## 6. Machine Learning Emerges (1990s–2000s)

### A Paradigm Shift

- Shifted from hand-coded rules to learning from data
- Computers learn patterns rather than being manually programmed

### Key Methods & Milestones

- New algorithms (such as SVMs) helped computers recognise patterns in data more accurately.
- Data mining emerged, focusing on discovering useful information from large datasets.
- In 1997, IBM's Deep Blue defeated chess champion Garry Kasparov, showing that machines could outperform humans in specific complex tasks.
- The growth of the internet generated vast amounts of data, enabling more effective training of machine learning algorithms.



Garry Kasparov and Deep Blue



## 7. Deep Learning Revolution (2010s)

### Enabling Factors

- Advances in computing power (GPUs enabled massive parallel processing)
- Availability of large datasets (ImageNet and others)
- Breakthroughs in neural network architectures (deep neural networks)

### Major Achievements

- ImageNet: In 2009, ImageNet project launched, led by Fei-Fei Li at Stanford. It is a large visual database of millions of labelled images used to train and evaluate computer vision systems.
- AlexNet (2012) – AlexNet was a deep learning model, trained on ImageNet to classify images, and it significantly outperformed previous computer vision methods.

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Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012).

ImageNet classification with deep convolutional neural networks.

Advances in Neural Information Processing Systems, 25

## 7. Deep Learning Revolution (2010s)

### Major Achievements - continued

- Speech recognition becoming practical (Siri, Google Assistant)
- Self-driving car research accelerating
- AlphaGo defeated world Go champion Lee Sedol (2016)
- Transformer architecture introduced (2017), which is the foundation of modern large language models and generative AI.



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Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I. (2017) 'Attention Is All You Need', Advances in Neural Information Processing Systems, 30.

## 8. Generative AI Era (2020s–Present)

### What is Generative AI

Generative AI is AI that creates new content—such as text, images, audio, or code—by learning patterns from large datasets.

- Conversational AI assistants - such as ChatGPT, Claude, Gemini
- Image generation systems - such as Stable Diffusion
- Coding assistant - such as GitHub Copilot

### The long-standing dream of AI is now becoming reality

- Medical diagnosis
- Customer service through chatbots
- Self-driving and autonomous transportation systems
- Software development

**Let's watch how AI continues to shape the future.**